Tutorial Sheet 1

1 Part A: Inference & Uncertainty quantification

Task 1

A single 1-hectare plot in a forest is surveyed and $Y = 80$ birds are counted. We can assume that the count $Y \sim$ Poisson(λ), where λ is the **expected number of birds per hectare**, so that $\hat{\lambda} = Y = 80$ and Var$(Y) = \lambda$. Derive a 95% Wald confidence interval for λ and comment.

Task 2

A great crested newt eDNA survey checks $n = 20$ ponds and detects the species at $y = 3$ of them. Assuming $Y \sim$ Binomial(n, p), where p is the probability occurrence with a MLE estimate $\hat{p} = y/n = 0.15$.

(a) Derive Var$(\hat{p})$ and compute a 95% (Wald) confidence interval for $\hat{p}$ (tip: recall that if $X \sim$ Bin(n, p); Var$(X) = np(1 - p)$.

(b) what do you notice about this interval?

Task 3

Occupancy models are widely used in ecology to model species presence while accounting for the fact that a species might present but undetected during a survey. Let z denote whether a site is truly occupied ($z = 1$) or not ($z = 0$), with **occupancy probability** Pr$(z = 1) = \psi$. Each site is visited on multiple occasions. On each visit we record a detection ($y = 1$) or non-detection ($y = 0$), and we detect the species — *if it is present* — with **detection probability** Pr$(y = 1 \mid z = 1) = p$. We assume that if the species is truly absent we cannot detect it, Pr$(y = 1 \mid z = 0) = 0$.

A site is surveyed J independent times and the species is **never detected**. A non-detection does not prove absence, so we quantify our belief that the site is occupied with the posterior probability Pr$(z = 1 \mid y_1 = 0, \ldots, y_J = 0)$.

(a) Write Pr(not detected in J visits $\mid z = 1$) and Pr(not detected in J visits $\mid z = 0$).

(b) Use the **law of total probability** to write Pr(not detected) (i.e., the probability that species is never detected)

(c) Use Bayes' theorem to obtain Pr$(z = 1 \mid y_1 = 0, \ldots, y_J = 0)$.

(d) A wetland is surveyed $J = 2$ times for a species of frog that was not detected in neither of the surveys. Suppose that we know from previous studies that $\psi = 0.8$ and $p = 0.5$. Compute the posterior probability that frog is present at that site.

Task 4

Several methods for dealing with values marked as being at the limit of detection within the paper by Eastoe et al (2006). Read the paper (available below), summarise the different methods compared by the authors and comment on what the conclusions were.

Task 5

(a) An ecologist wishes to analyse a dataset that contains a variable with around 1% of its data censored at a limit of detection. The ecologist proposes to use a simple substitution method to replace all values at the limit of detection (c_L) with $0.5c_L$. Do you agree with this approach? Why/ why not?

(b) The ecologist wishes to analyse another dataset that contains a variable with around 55% of the data censored at a limit of detection. The ecologist would like to take the same simple substitution approach as in part (a). Do you agree with this approach? What advice would you give?

Task 6

(a) Suppose that chlorophyll data in a freshwater loch are collected twice a week, but the equipment needs to be removed for maintenance once a month, so that there are around 12 missing values per year. Can we assume that these values are missing at random?

(b) Suppose that in another loch, the data are also collected twice a week, but the monitoring device there only needs to be maintained once a year. If this is removed every December, so that there are no values for that month, can we assume that these data are missing at random? Might we need to impute the data?

Level M Extra Question

A pilot survey at a comparable site detected the newt in **2 of 10** ponds. Use this as prior information in a Bayesian analysis and update it based on the information from the previous task (i.e., $y = 3$ detections in $n = 20$ ponds). To do so follow the next steps:

1. Write the likelihood of the pilot study and show it has the form of a Beta distribution where the general Beta density $\pi(p) \propto p^{a-1}(1-p)^{b-1}$.
2. Identify the likelihood and the prior for the main survey (recall that we assume the main survey follows a binomial density such that $\pi(y \mid p) \propto p^y(1-p)^{n-y}$ for $y = 3$ and $n = 20$.)
3. Derive the posterior distribution by multiplying the likelihood by the prior. What family results?
4. Compute the posterior mean of p using the fact that if $X \sim \text{Beta}(a,b)$; $\mathbb{E}(X) = \frac{a}{a+b}$)
5. Compute a 95% credible intervals for p and comment (you can use R to compute the quantiles directly)

2 Part B: Sampling and monitoring

University of Glasgow

Note

This part of the tutorial sheet relates to material in Week 3, which we'll cover in the lectures during the same week as Tutorial 1. You can attempt these questions in Tutorial 1 if you wish, but you may prefer to go through these during Tutorial 2 instead.

Task 7

In the case of a simple random sample $x_1, \ldots, x_n$ of a random variable, X, assuming the observations are independent,

(a) derive the expected value of X^2
(b) derive the expected value of $\bar{X}^2$
(c) show that the sample variance, s^2, is an unbiased estimator of the population variance, σ^2.

Task 8

Gilbert (1977) reports results of soil sampling at a nuclear weapons test area obtained using stratified random sampling to assess the total amount of Plutonium found in surface soil. Use the information in the table below to:

(a) Estimate the total inventory and derive the estimator for the variance of the totals

(b) Determine the optimal number of population units of measure in each of the 4 strata. Find the total number of units to sample, assuming cost is fixed (where total=£50,000 and cost per unit is £500 for all strata).

strata	Size $\times$ area of the stratum N_l	n_l	Mean for stratum	Variance s_l^2
1	351,000	18	4.1	30.42
2	82,300	12	73	10,800
3	26,200	13	270	127,413
4	11,000	20	260	84,500

Tip

The stratified estimator of the population total is

$$\hat{I} = \sum_{l=1}^{L} N_l \, \bar{y}_l.$$

You may use the fact that the variance for the mean of l-th strata is given by

$$\text{Var}(\bar{y}_l) = \left(1 - \frac{n_l}{N_l}\right) \frac{s_l^2}{n_l}$$

Task 9

Discuss the advantages and disadvantages of the three sampling methods below for mapping a pollutant field:

- Simple random sampling
- Systematic sampling
- Stratified random sampling

Task 10

The Water Framework Directive states:

"Member states must ensure that enough individual water bodies of each water type are monitored and determine how many stations are required to determine the ecological and chemical status of the water body"

Discuss briefly how you would translate this statement into a monitoring programme, given that there are 6 different water body types comprising 10%, 25%, 30%, 20%, 10% and 5% of the total population of 6600 water bodies and that your limited resources only allow you to study a total of 200 water bodies. Knowledge of the within-type variability is not available.

Task 11

Read pages 17–23 of the Analytical Laboratories for the Measurement of Environmental Radioactivity (ALMERA) report on soil sampling (available below). Describe briefly the sampling strategy adopted and also the methods of analysis presented.

Task 12

Read the SEPA survey of business waste (available below) and describe briefly the sampling strategy you would propose. Discuss its advantages and disadvantages.